

Step 3 Draft

Project Group 7

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## Cookbook Central

URL: <http://classwork.engr.oregonstate.edu:62472/>

### Executive Summary

Design changes we decided to make based on feedback are the following:

- Updated our overview since we weren't specific enough and lacked numbers
- Renamed the attribute food\_type in the table to meal\_type, to make it more clear that we're referring to breakfast/lunch/dinner rather than diet type
- Created a followers table so a user could follow other users, since it was suggested that our website could act like a social media website for recipe sharing
- Omitted 'NULL' from attributes in DB outline, and removed the 'unique' constant from primary keys. Made primary keys auto-incrementing in intersection tables.
- Added CASCADE operations to the SQL file for data definition queries, to preserve referential integrity when a primary key is deleted from a table.
- Added stars to the Reviews page, instead of having it display an integer from 1 through 5
- Added type to email input, to ensure that the email address contains an '@'

Design changes we decided to make on our own are the following:

- Specified that userId in Passwords will be auto-incremented
- Changed the data type of totalCost in Recipes and cost in Ingredients from int to float, since we need to include cents
- Changed the data type of quantity in RecipeIngredients from int to varchar, since we realized that users would need to be able to input fractions
- Made the quantity in RecipeIngredients 'not NULL', since it's mandatory for users to specify how much of each ingredient is in their recipe
- Removed the Passwords table, since it's not a customer facing website so there's no need for a login page

## Project Outline

The biggest problem with physical cookbooks are their expensive costs, they can easily be lost or damaged, and they are hard to share with friends. On average, most households purchase 2–3 cookbooks a year, which can become costly because cookbook prices can range from \$20 up to over \$100 for more specialized cooking. Our database driven website will help users keep a free digital cookbook, which will provide them with an easy-to-access place to store all their favorite recipes, and prevent recipes from being lost. Users will be able to post their own recipes, view other users' recipes, and leave reviews on any recipes. They'll also be able to save their own recipes or other users' recipes to their cookbook, allowing the user to have access to dozens of their favorite recipes. Additionally, users will be able to follow each other. Approximately 50 recipes will be held in the database, and 10 users. We estimate there will be no more than 100 unique ingredients. Each user will only be able to post one review on each recipe.

## Database Outline

Entities:

- **Users:** Records information about the users using our website.
  - **userID:** int, auto\_increment, not NULL, PK
  - **firstName:** varchar, not NULL
  - **lastName:** varchar, not NULL
  - **email:** varchar, not NULL
  - **Relationship:**
    - A 1:M relationship between Users and Reviews is implemented with userID as a FK inside of Reviews.
    - A 1:M relationship between Users and Recipes is implemented with userID as a FK inside of Recipes.
    - A M:N relationship between Users and Recipes is implemented with userID as a FK inside an intersection table named SavedRecipes.
    - A M:N relationship between Users and itself is implemented with userID as a FK inside an intersection table named UserFollowers.
- **Recipes:** Stores information about recipes.
  - **recipeID:** int, auto\_increment, not NULL, PK
  - **userID:** int, FK
  - **name:** varchar, not NULL
  - **mealType:** varchar
  - **duration:** int
  - **servingSize:** int
  - **totalCost:** float
  - **directions:** varchar, not NULL

- **Relationship:**
  - A 1:M relationship between Recipes and Reviews is implemented with recipeID as a FK inside of Reviews.
  - A M:N relationship between Recipes and Users is implemented with recipeID as a FK inside an intersection table named SavedRecipes.
  - A M:N relationship between Recipes and Ingredients is implemented with recipeID as a FK inside an intersection table named RecipeIngredients.
- **Ingredients:** Stores information about ingredients used in recipes.
  - **ingredientID:** int, auto\_increment, not NULL, PK
  - **name:** varchar, not NULL
  - **cost:** float
  - **Relationship:**
    - A M:N relationship between Ingredients and Recipes is implemented with ingredientID as a FK inside an intersection table named RecipeIngredients.
- **Reviews:** Stores information about reviews left on recipes
  - **reviewID:** int, auto\_increment, not NULL, PK
  - **recipeID:** int, FK
  - **userID:** int, FK
  - **date:** date, not NULL
  - **rating:** int, not NULL
  - **comment:** varchar
  - **Relationship:**
    - A M:1 relationship between Reviews and Recipes is implemented with recipeID as a FK inside of Reviews.
    - A M:1 relationship between Reviews and Users is implemented with userID as a FK inside of Reviews.

Intersection Entity:

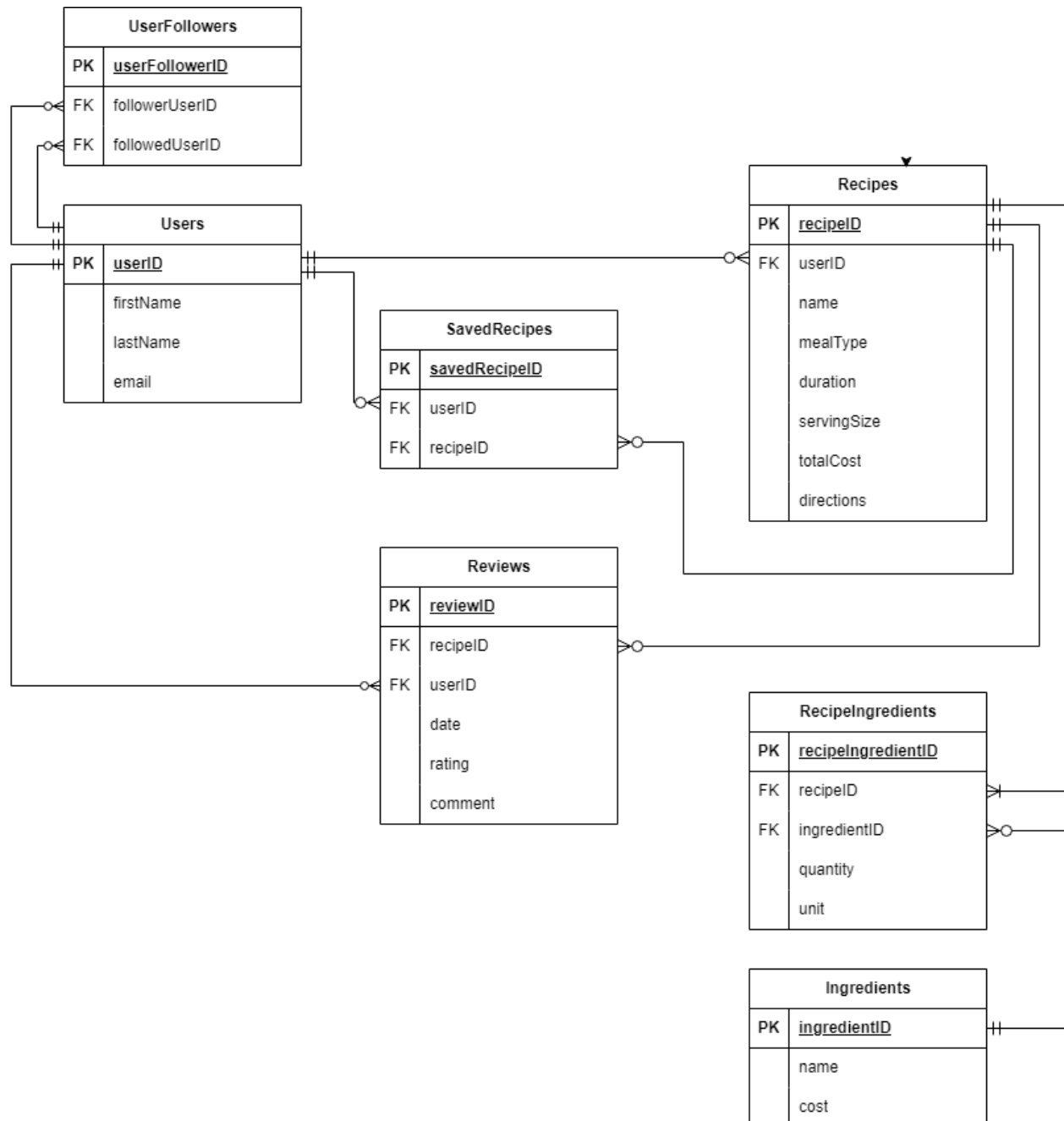
- **RecipeIngredients:** Stores information about the quantity and unit for a specific recipe's ingredient, connecting the Recipes and Ingredients entities.
  - **recipeIngredientID:** int, auto\_increment, not NULL, PK
  - **recipeID:** int, FK
  - **ingredientID:** int, FK
  - **quantity:** varchar, not NULL
  - **unit:** varchar

Intersection Tables:

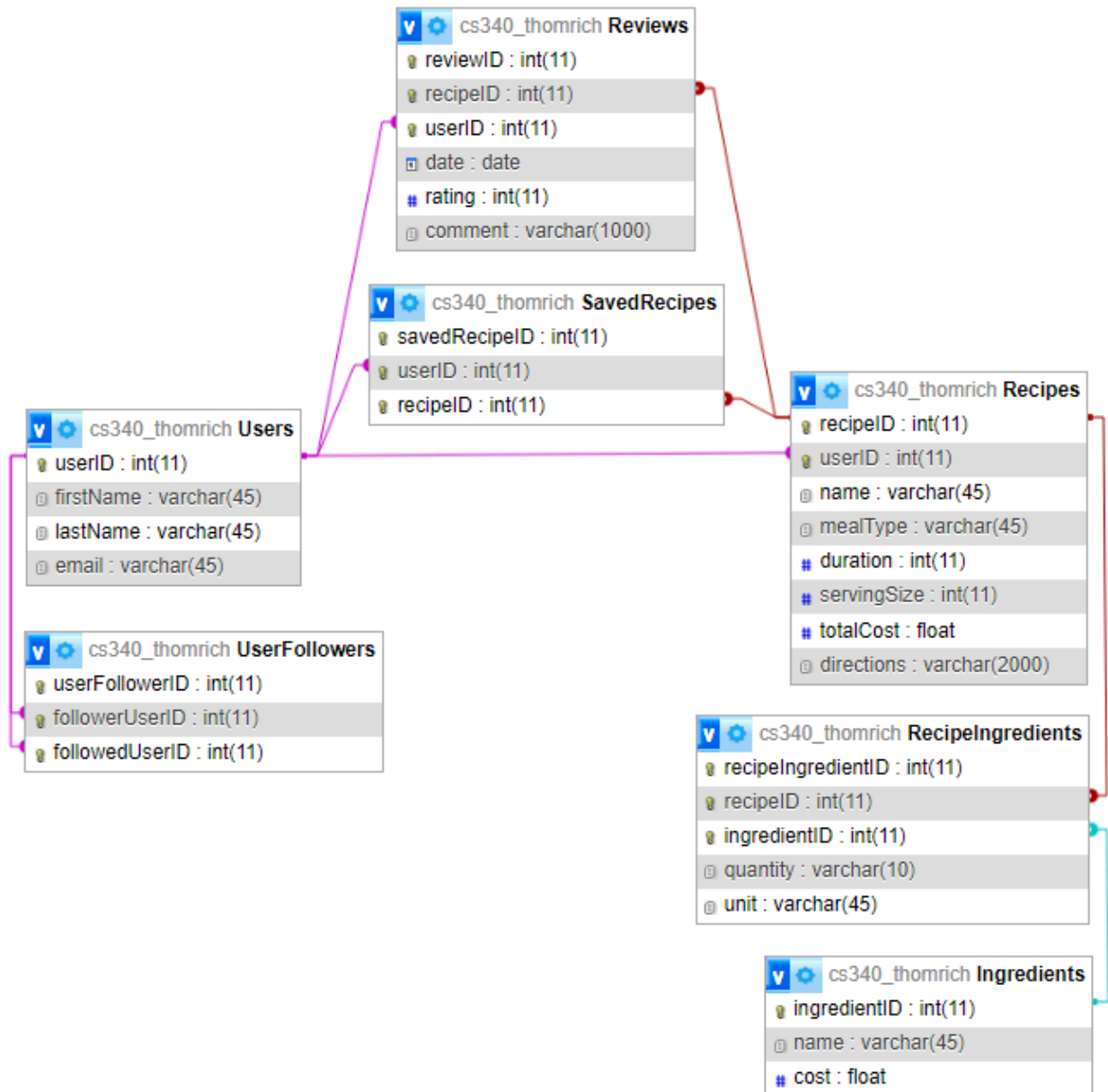
- **SavedRecipes:** Connects the user entity to the recipe entity.
  - **savedRecipeID:** int, auto\_increment, not NULL, PK

- **userID**: int, FK
- **recipeID**: int, FK
- **UserFollowers**: Stores information on what users each user follows.
  - **userFollowerID**: int, auto\_increment, not NULL, PK
  - **followerUserID**: int, FK
  - **followedUserID**: int, FK

## Entity Relationship Diagram



## Schema



## Sample Data

### Users

| userID | firstName | lastName  | email                    |
|--------|-----------|-----------|--------------------------|
| 1      | Michael   | Sams      | michaelsams247@gmail.com |
| 2      | John      | Doe       | johndoe@gmail.com        |
| 3      | Kenny     | Powers    | kennypowers@gmail.com    |
| 4      | Leslie    | Knope     | leslieknope@gmail.com    |
| 5      | Tom       | Haverford | thaverford@gmail.com     |

### UserFollowers

| userFollowerID | followerUserID | followedUserID |
|----------------|----------------|----------------|
| 1              | 1              | 2              |
| 2              | 1              | 3              |
| 3              | 2              | 3              |
| 4              | 4              | 1              |
| 5              | 4              | 5              |

### SavedRecipes

| savedRecipeID | userID | recipeID |
|---------------|--------|----------|
| 1             | 1      | 2        |
| 2             | 2      | 1        |
| 3             | 3      | 1        |
| 4             | 4      | 4        |
| 5             | 4      | 2        |

## Reviews

| reviewID | recipeID | userID | date       | rating | comment                                              |
|----------|----------|--------|------------|--------|------------------------------------------------------|
| 1        | 2        | 1      | 2024-06-07 | 3      | So tasty!                                            |
| 2        | 1        | 2      | 2024-07-07 | 4      | My family and I loved this dish! So flavorful.       |
| 3        | 1        | 3      | 2024-07-11 | 1      | Horrible. If I could rate it less than a 1, I would. |
| 4        | 3        | 5      | 2024-07-13 | 5      | Best waffles ever!!!                                 |
| 5        | 2        | 5      | 2024-07-14 | 2      | NULL                                                 |

## Recipes

| recipeID | userID | name                    | mealType  | duration | servingSize | totalCost | directions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|--------|-------------------------|-----------|----------|-------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 1      | Honey Garlic Pork Chops | Dinner    | 45       | 4           | NULL      | Step 1: Season pork chops with salt, pepper, and garlic powder. Step 2: In a pan on medium heat, add oil and cook each side of the pork chops for 4–5 minutes, or until done. Then remove pork chops from pan. Step 3: In that same pan add in butter and garlic and sauté until fragrant. Step 4: Turn up heat to medium high and add in honey, apple cider vinegar and water. Step 5: Once the sauce has thickened, add back in the pork chops and baste the sauce over the pork chops for 2 minutes. Step 6: Remove from heat and serve. |
| 2        | 2      | Roasted Potatoes        | Side      | 45       | 4           | NULL      | Step 1: Cut mini potatoes into 1/4ths. Step 2: Cover potatoes in olive oil. Combine seasoning in bowl and then toss in the potatoes. Step 3: Bake in oven at 425 degree F for 30-40 minutes.                                                                                                                                                                                                                                                                                                                                                |
| 3        | 4      | Matcha Waffles          | Breakfast | 30       | 8           | NULL      | Step 1: Beat eggs in a large mixing bowl, then gradually add the rest of the ingredients in order except for the chocolate chips. Step 2: Heat the waffle iron. Pour batter in the iron and sprinkle several chocolate chips on top, spoon a little more batter on top to cover the chocolate chips. Cover the lid and bake for 2.5 minutes. Step 3: Remove the waffle from the iron and cool on a rack for at least 1 minute.                                                                                                              |

|   |   |                                        |         |    |    |      |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---|---|----------------------------------------|---------|----|----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | 4 | Double<br>Chocolate<br>Chip<br>Cookies | Dessert | 40 | 18 | NULL | Step 1: Preheat an oven to 350 degrees F. Line parchment paper on a baking sheet. Step 2: In a large mixing bowl, stir butter and both sugars until combined. Add the egg and stir. Step 3: Add vanilla, salt, cocoa powder, flour, and baking soda to the bowl and stir until combined. Step 4: Fold in chocolate chips. Step 5: Use a cookie scoop to place the dough on the baking sheet, then bake for 12 minutes. |
|---|---|----------------------------------------|---------|----|----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### RecipeIngredients

| recipeIngredientID | recipeID | ingredientID | quantity | unit      |
|--------------------|----------|--------------|----------|-----------|
| 1                  | 1        | 1            | 1/4      | cup       |
| 2                  | 1        | 2            | 4        | NULL      |
| 3                  | 1        | 3            | 8        | cloves    |
| 4                  | 1        | 4            | 1        | to season |
| 5                  | 1        | 5            | 1        | to season |
| 6                  | 1        | 6            | 2        | Tbsp      |
| 7                  | 1        | 7            | 1        | to season |
| 8                  | 1        | 8            | 1        | Tbsp      |
| 9                  | 1        | 9            | 2        | Tbsp      |
| 10                 | 2        | 4            | 1        | tsp       |
| 11                 | 2        | 5            | 1        | tsp       |
| 12                 | 2        | 7            | 2        | tsp       |
| 13                 | 2        | 9            | 1        | Tbsp      |
| 14                 | 2        | 10           | 1        | tsp       |
| 15                 | 2        | 11           | 2        | tsp       |
| 16                 | 2        | 12           | 1        | Tbsp      |
| 17                 | 2        | 13           | 1.5      | lb        |
| 18                 | 2        | 14           | 1/4      | cup       |
| 19                 | 3        | 15           | 2        | whole     |
| 20                 | 3        | 16           | 1        | cup       |
| 21                 | 3        | 17           | 3/4      | cup       |

|    |   |    |       |       |
|----|---|----|-------|-------|
| 22 | 3 | 18 | 3/4   | cup   |
| 23 | 3 | 19 | 2     | Tbsp  |
| 24 | 3 | 21 | 1/2   | cup   |
| 25 | 3 | 23 | 1/4   | cup   |
| 26 | 3 | 24 | 1     | Tbsp  |
| 27 | 3 | 25 | 2     | Tbsp  |
| 28 | 3 | 26 | 1     | Tbsp  |
| 29 | 3 | 9  | 2     | Tbsp  |
| 30 | 3 | 22 | 1/2   | cup   |
| 31 | 3 | 27 | 1/2   | tsp   |
| 32 | 3 | 28 | 1 1/2 | tsp   |
| 33 | 3 | 29 | 1/3   | cup   |
| 34 | 4 | 8  | 1     | stick |
| 35 | 4 | 19 | 1/3   | cup   |
| 36 | 4 | 20 | 1/3   | cup   |
| 37 | 4 | 15 | 1     | whole |
| 38 | 4 | 30 | 1     | tsp   |
| 39 | 4 | 4  | 1     | pinch |
| 40 | 4 | 31 | 1/4   | cup   |
| 41 | 4 | 22 | 1 1/4 | cup   |
| 42 | 4 | 27 | 1/2   | tsp   |
| 43 | 4 | 29 | 1 1/8 | cup   |

## Ingredients

| ingredientID | name                     | cost |
|--------------|--------------------------|------|
| 1            | Honey                    | NULL |
| 2            | Pork Chops               | NULL |
| 3            | Garlic                   | NULL |
| 4            | Salt                     | NULL |
| 5            | Pepper                   | NULL |
| 6            | Apple Cider Vinegar      | NULL |
| 7            | Garlic Powder            | NULL |
| 8            | Unsalted Butter          | NULL |
| 9            | Olive Oil                | NULL |
| 10           | Onion Powder             | NULL |
| 11           | Paprika                  | NULL |
| 12           | Italian Seasoning        | NULL |
| 13           | Baby Potatoes            | NULL |
| 14           | Water                    | NULL |
| 15           | Egg                      | NULL |
| 16           | Buttermilk               | NULL |
| 17           | Milk                     | NULL |
| 18           | Rolled Oats              | NULL |
| 19           | White Sugar              | NULL |
| 20           | Brown Sugar              | NULL |
| 21           | Almond Flour             | NULL |
| 22           | All-Purpose Flour        | NULL |
| 23           | Flaxseed Meal            | NULL |
| 24           | Chia Seeds               | NULL |
| 25           | Hemp Seeds               | NULL |
| 26           | Matcha Powder            | NULL |
| 27           | Baking Soda              | NULL |
| 28           | Baking Powder            | NULL |
| 29           | Chocolate Chips          | NULL |
| 30           | Vanilla Extract          | NULL |
| 31           | Unsweetened Cocoa Powder | NULL |

# UI Screenshots

**Homepage:** (Click “Cookbook Central” in the navigation bar to get back to this home page)

| Table Name         | Description                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recipes            | An entity table that contains data about recipes.                                                                                                                     |
| Recipe Ingredients | An intersection table that supports the M:N relationship between Recipes and Ingredients. This table contains data about which ingredients are used in which recipes. |
| Ingredients        | An entity table that contains data about ingredients.                                                                                                                 |
| Saved Recipes      | An intersection table that supports the M:N relationship between Users and Recipes. This table contains data about which recipes are saved by which users.            |
| Reviews            | An entity table that contains data about reviews posted by users on recipes.                                                                                          |
| Users              | An entity table that contains data about users.                                                                                                                       |
| User Followers     | An intersection table that supports the M:N relationship between Users and Users. This table contains data about the follower-followed relationship between users.    |

## Recipes: READ/DELETE

Clicking a red “Delete” button on the Recipes page will delete the corresponding row in the table. Delete from a M:N relationship is implemented here, as deleting a recipe will delete any rows in Recipe Ingredients, Saved Recipes, and Reviews that have a reference to that recipe.

| Recipe ID | User Name    | Name                          | Meal Type | Duration | Serving Size | Total Cost | Directions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                             |
|-----------|--------------|-------------------------------|-----------|----------|--------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 1         | Michael Sams | Honey Garlic Pork Chops       | Dinner    | 45       | 4            | None       | Step 1: Season pork chops with salt, pepper, and garlic powder. Step 2: In a pan on medium heat, add oil and cook each side of the pork chops for 4-5 minutes, or until done. Then remove pork chops from pan. Step 3: In that same pan add in butter and garlic and sauté until fragrant. Step 4: Turn up heat to medium high and add in honey, apple cider vinegar and water. Step 5: Once the sauce has thickened, add back in the pork chops and baste the sauce over the pork chops for 2 minutes. Step 6: Remove from heat and serve. | <a href="#">Edit</a> <a href="#">Delete</a> |
| 2         | John Doe     | Roasted Potatoes              | Side      | 45       | 4            | None       | Step 1: Cut mini potatoes into 1/4ths. Step 2: Cover potatoes in olive oil. Combine seasoning in bowl and then toss in the potatoes. Step 3: Bake in oven at 425 degree F for 30-40 minutes.                                                                                                                                                                                                                                                                                                                                                | <a href="#">Edit</a> <a href="#">Delete</a> |
| 3         | Leslie Knope | Matcha Waffles                | Breakfast | 30       | 8            | None       | Step 1: Beat eggs in a large mixing bowl, then gradually add the rest of the ingredients in order except for the chocolate chips. Step 2: Heat the waffle iron. Pour batter in the iron and sprinkle several chocolate chips on top, spoon a little more batter on top to cover the chocolate chips. Cover the lid and bake for 2.5 minutes. Step 3: Remove the waffle from the iron and cool on a rack for at least 1 minute.                                                                                                              | <a href="#">Edit</a> <a href="#">Delete</a> |
| 4         | Leslie Knope | Double Chocolate Chip Cookies | Dessert   | 40       | 18           | None       | Step 1: Preheat an oven to 350 degrees F. Line parchment paper on a baking sheet. Step 2: In a large mixing bowl, stir butter and both sugars until combined. Add the egg and stir. Step 3: Add vanilla, salt, cocoa powder, flour, and baking soda to the bowl and stir until combined. Step 4: Fold in chocolate chips. Step 5: Use a cookie scoop to place the dough on the baking sheet, then bake for 12 minutes.                                                                                                                      | <a href="#">Edit</a> <a href="#">Delete</a> |

## Recipes: CREATE

Clicking the green “Add” button on the Recipes page takes the user to this page, where they can select a user from a drop-down menu and add data for the new recipe.

| User Name    | Name | MealType | Duration | ServingSize | TotalCost | Directions |
|--------------|------|----------|----------|-------------|-----------|------------|
| Michael Sams |      |          |          |             |           |            |

## Recipes: UPDATE

Clicking a yellow “Edit” button on the Recipes page takes the user to this page, where fields are pre-populated with data from the corresponding row in the table.

| User Name    | Name                    | MealType | Duration | ServingSize | TotalCost | Directions                  |
|--------------|-------------------------|----------|----------|-------------|-----------|-----------------------------|
| Michael Sams | Honey Garlic Pork Chops | Dinner   | 45       | 4           | None      | Step 1: Season pork chops v |

## Recipe Ingredients: READ/DELETE

Clicking a red “Delete” button on the Recipe Ingredients page will delete the corresponding row in the table.

| Recipe Ingredient ID | Recipe Name             | Ingredient Name     | Quantity | Unit      | Add         |
|----------------------|-------------------------|---------------------|----------|-----------|-------------|
| 1                    | Honey Garlic Pork Chops | Honey               | 1/4      | Cup       | Edit Delete |
| 2                    | Honey Garlic Pork Chops | Pork Chops          | 4        | None      | Edit Delete |
| 3                    | Honey Garlic Pork Chops | Garlic              | 8        | Cloves    | Edit Delete |
| 4                    | Honey Garlic Pork Chops | Salt                | 1        | To season | Edit Delete |
| 5                    | Honey Garlic Pork Chops | Pepper              | 1        | To season | Edit Delete |
| 6                    | Honey Garlic Pork Chops | Apple Cider Vinegar | 2        | Tbsp      | Edit Delete |
| 7                    | Honey Garlic Pork Chops | Garlic Powder       | 1        | To season | Edit Delete |
| 8                    | Honey Garlic Pork Chops | Unsalted Butter     | 1        | Tbsp      | Edit Delete |
| 9                    | Honey Garlic Pork Chops | Olive Oil           | 2        | Tbsp      | Edit Delete |

## Recipe Ingredients: CREATE

Clicking the green “Add” button on the Recipe Ingredients page takes the user to this page, where they can select a recipe and ingredient from a drop-down menu and add data for the new recipe ingredient.

| Recipe Name             | Ingredient Name | Quantity | Unit |
|-------------------------|-----------------|----------|------|
| Honey Garlic Pork Chops | Honey           |          |      |

## Recipe Ingredients: UPDATE

Clicking a yellow “Edit” button on the Recipe Ingredients page takes the user to this page, where fields are pre-populated with data from the corresponding row in the table. **Update of a M:N relationship is implemented here, since foreign keys can be changed to reference a different primary key by selecting a different recipe name or different ingredient name from the drop-down menu.**

127.0.0.1:5000/edit/RecipeIngredients

**Cookbook Central** Recipes Recipe Ingredients Ingredients Saved Recipes Reviews Users User Followers

Recipe Name Ingredient Name Quantity Unit

Honey Garlic Pork Chops Honey 1/4 Cup **Update**

## Ingredients: READ

127.0.0.1:5000/ingredients

**Cookbook Central** Recipes Recipe Ingredients Ingredients Saved Recipes Reviews Users User Followers

| Ingredient ID | Name                | Cost |                           |
|---------------|---------------------|------|---------------------------|
| 1             | Honey               | None | <b>Add</b><br><b>Edit</b> |
| 2             | Pork Chops          | None | <b>Edit</b>               |
| 3             | Garlic              | None | <b>Edit</b>               |
| 4             | Salt                | None | <b>Edit</b>               |
| 5             | Pepper              | None | <b>Edit</b>               |
| 6             | Apple Cider Vinegar | None | <b>Edit</b>               |

## Ingredients: CREATE

Clicking the green “Add” button on the Ingredients page takes the user to this page, where they can add data for the new ingredient.

127.0.0.1:5000/insert/Ingredients

**Cookbook Central** Recipes Recipe Ingredients Ingredients Saved Recipes Reviews Users User Followers

Name Cost

**Insert**

## Ingredients: UPDATE

Clicking a yellow “Edit” button on the Ingredients page takes the user to this page, where fields are pre-populated with data from the corresponding row in the table.

127.0.0.1:5000/edit/Ingredients

**Cookbook Central** Recipes Recipe Ingredients Ingredients Saved Recipes Reviews Users User Followers

Name Cost

Honey None **Update**

**Saved Recipes: READ/DELETE**

Clicking a red “Delete” button on the Saved Recipes page will delete the corresponding row in the table.

127.0.0.1:5000/savedRecipes

Cookbook Central

Recipes

Recipe Ingredients

Ingredients

Saved Recipes

Reviews

Users

User Followers

| Saved Recipe ID | User Name    | Recipe Name                   |                 |                   |
|-----------------|--------------|-------------------------------|-----------------|-------------------|
| 2               | John Doe     | Honey Garlic Pork Chops       | <div>Edit</div> | <div>Delete</div> |
| 3               | Kenny Powers | Honey Garlic Pork Chops       | <div>Edit</div> | <div>Delete</div> |
| 1               | Michael Sams | Roasted Potatoes              | <div>Edit</div> | <div>Delete</div> |
| 4               | Leslie Knope | Roasted Potatoes              | <div>Edit</div> | <div>Delete</div> |
| 5               | Leslie Knope | Double Chocolate Chip Cookies | <div>Edit</div> | <div>Delete</div> |

**Saved Recipes: CREATE**

Clicking the green “Add” button on the Saved Recipes page takes the user to this page, where they can select a user and recipe from a drop-down menu and add data for the new saved recipe.

127.0.0.1:5000/insert/SavedRecipes

Cookbook Central

Recipes

Recipe Ingredients

Ingredients

Saved Recipes

Reviews

Users

User Followers

User Name

Recipe Name

Michael Sams

Honey Garlic Pork Chops

Insert

**Saved Recipes: UPDATE**

Clicking a yellow “Edit” button on the Saved Recipes page takes the user to this page, where fields are pre-populated with data from the corresponding row in the table.

127.0.0.1:5000/edit/SavedRecipes

Cookbook Central

Recipes

Recipe Ingredients

Ingredients

Saved Recipes

Reviews

Users

User Followers

User Name

Recipe Name

Michael Sams

Roasted Potatoes

Update

**Reviews: READ/DELETE**

Clicking a red “Delete” button on the Reviews page will delete the corresponding row in the table.

127.0.0.1:5000/reviews

Cookbook Central

Recipes

Recipe Ingredients

Ingredients

Saved Recipes

Reviews

Users

User Followers

| Review ID | Recipe Name             | User Name     | Date       | Rating | Comment                                              |                 |                   |
|-----------|-------------------------|---------------|------------|--------|------------------------------------------------------|-----------------|-------------------|
| 2         | Honey Garlic Pork Chops | John Doe      | 2024-07-07 | ★★★★★  | My family and I loved this dish! So flavorful.       | <div>Edit</div> | <div>Delete</div> |
| 3         | Honey Garlic Pork Chops | Kenny Powers  | 2024-07-11 | ★★★★★  | Horrible. If I could rate it less than a 1, I would. | <div>Edit</div> | <div>Delete</div> |
| 1         | Roasted Potatoes        | Michael Sams  | 2024-06-07 | ★★★★★  | So tasty!                                            | <div>Edit</div> | <div>Delete</div> |
| 5         | Roasted Potatoes        | Tom Haverford | 2024-07-14 | ★★★★★  | None                                                 | <div>Edit</div> | <div>Delete</div> |
| 4         | Matcha Waffles          | Tom Haverford | 2024-07-13 | ★★★★★  | Best waffles ever!!!                                 | <div>Edit</div> | <div>Delete</div> |

## Reviews: CREATE

Clicking the green “Add” button on the Reviews page takes the user to this page, where they can select a recipe and user from a drop-down menu and add data for the new review.

127.0.0.1:5000/insert/Reviews

**Cookbook Central** Recipes Recipe Ingredients Ingredients Saved Recipes Reviews Users User Followers

Recipe Name User Name Date Rating Comment

Honey Garlic Pork Chops Michael Sams mm / dd / yyyy

Insert

## Reviews: UPDATE

Clicking a yellow “Edit” button on the Reviews page takes the user to this page, where fields are pre-populated with data from the corresponding row in the table.

A NULLable relationship is implemented here, since the user can be set to null. Also, if a user is deleted, the review would stay but the foreign key for the user would be set to NULL.

127.0.0.1:5000/edit/Reviews

**Cookbook Central** Recipes Recipe Ingredients Ingredients Saved Recipes Reviews Users User Followers

Recipe Name User Name Date Rating Comment

Honey Garlic Pork Chops John Doe 07 / 07 / 2024 4 My family and I loved this dis

Update

## Users: READ/DELETE

Clicking a red “Delete” button on the Users page will delete the corresponding row in the table.

127.0.0.1:5000/users

**Cookbook Central** Recipes Recipe Ingredients Ingredients Saved Recipes Reviews Users User Followers

| User ID | First Name | Last Name | Email                    |      |        |
|---------|------------|-----------|--------------------------|------|--------|
| 1       | Michael    | Sams      | michaelsams247@gmail.com | Edit | Delete |
| 2       | John       | Doe       | johndoe@gmail.com        | Edit | Delete |
| 3       | Kenny      | Powers    | kennypowers@gmail.com    | Edit | Delete |
| 4       | Leslie     | Knope     | leslieknope@gmail.com    | Edit | Delete |
| 5       | Tom        | Haverford | thaverford@gmail.com     | Edit | Delete |

Add

## Users: CREATE

Clicking the green “Add” button on the Users page takes the user to this page, where they can add data for the new user.

127.0.0.1:5000/insert/Users

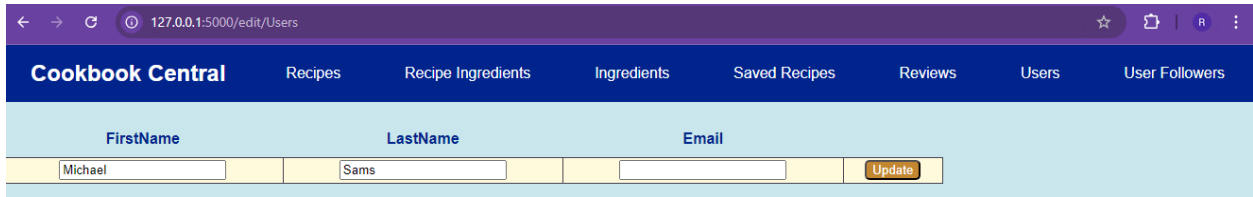
**Cookbook Central** Recipes Recipe Ingredients Ingredients Saved Recipes Reviews Users User Followers

FirstName LastName Email

Insert

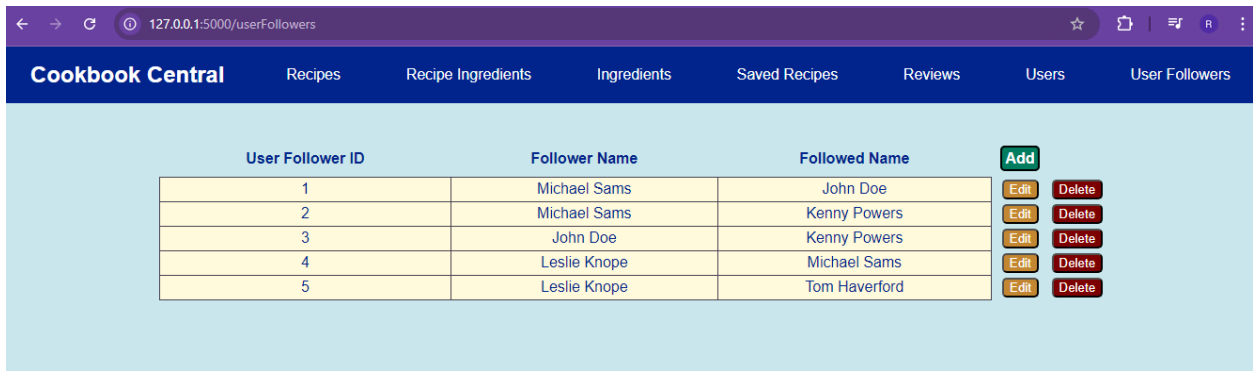
## Users: UPDATE

Clicking a yellow “Edit” button on the Users page takes the user to this page, where fields are pre-populated with data from the corresponding row in the table.



## User Followers: READ/DELETE

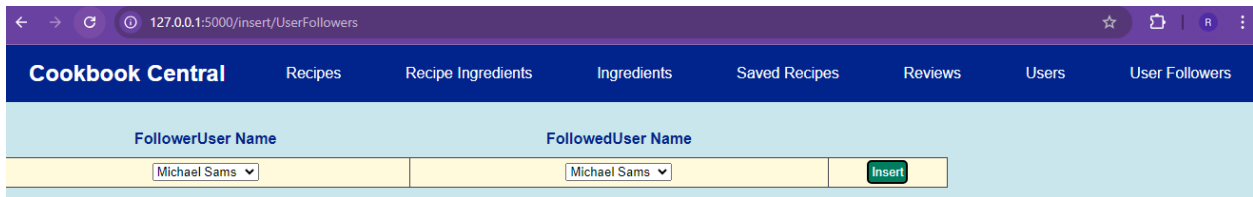
Clicking a red “Delete” button on the User Followers page will delete the corresponding row in the table.



| User Follower ID | Follower Name | Followed Name |      |        |
|------------------|---------------|---------------|------|--------|
| 1                | Michael Sams  | John Doe      | Edit | Delete |
| 2                | Michael Sams  | Kenny Powers  | Edit | Delete |
| 3                | John Doe      | Kenny Powers  | Edit | Delete |
| 4                | Leslie Knope  | Michael Sams  | Edit | Delete |
| 5                | Leslie Knope  | Tom Haverford | Edit | Delete |

## User Followers: CREATE

Clicking the green “Add” button on the User Followers page takes the user to this page, where they can select two users from a drop-down menu and add data for the new relationship.



## User Followers: UPDATE

Clicking a yellow “Edit” button on the User Followers page takes the user to this page, where fields are pre-populated with data from the corresponding row in the table.

